MONO SINGLE-ENDED CLASS-A | 40 V SINGLE SUPPLY / CAP-COUPLED | 8 ohm / 12-15 W target

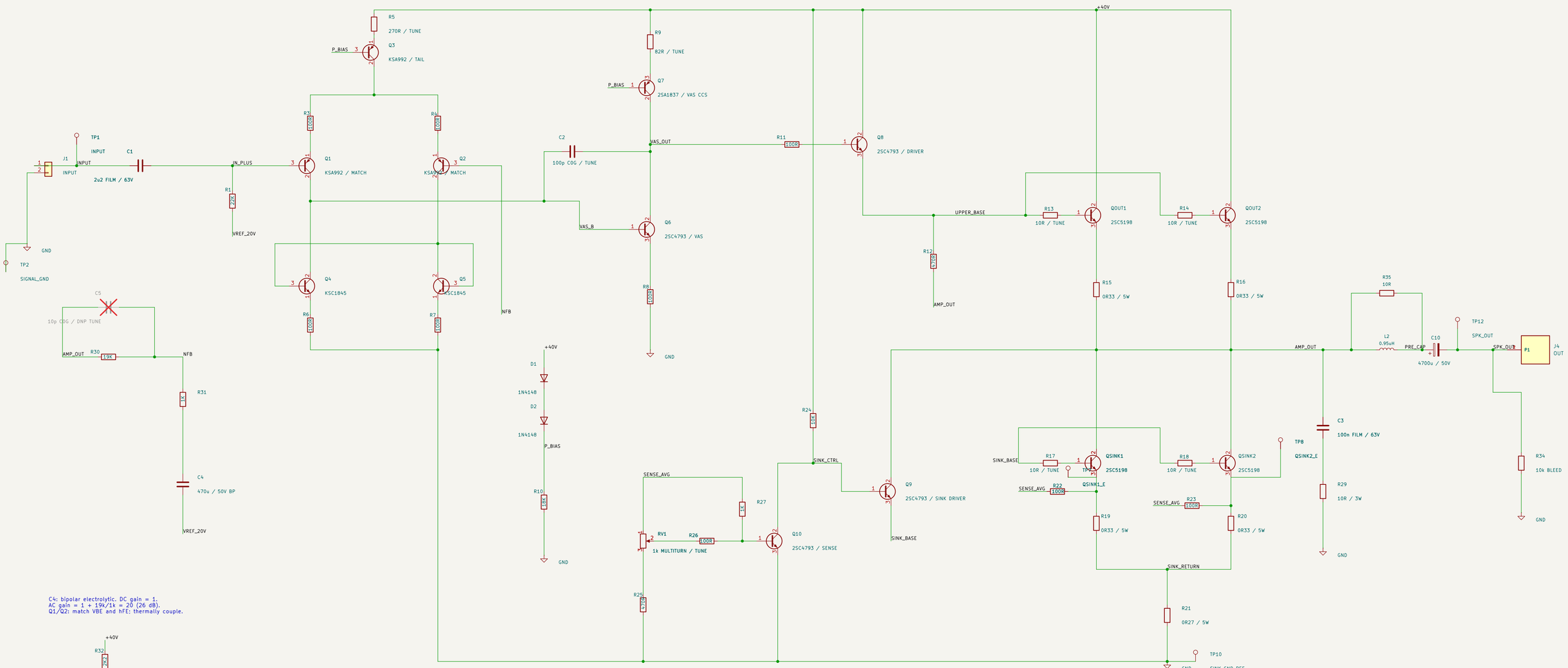
INPUT / DIFFERENTIAL PAIR

CURRENT MIRROR / VAS

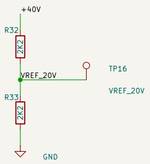
UPPER SIGNAL DRIVER

NPN CLASS-A POWER STAGE

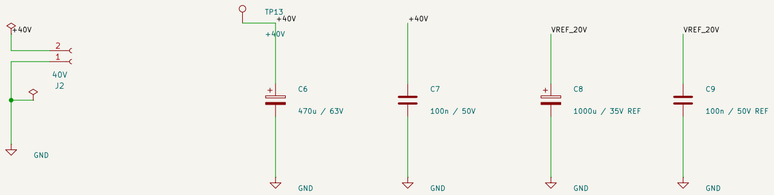
OUTPUT / SPEAKER



C4: bipolar electrolytic. DC gain = 1.
AC gain = 1 + 19k/1k = 20 (26 dB).
Q1/Q2: match VBE and hFE; thermally couple.



EXTERNAL 40 V SUPPLY / LOCAL DECOUPLING



GROUND: one GND net. Separate input/feedback returns from speaker/Zobel and rail-cap returns.
Join at J3 GND/local capacitor star. VREF is a quiet signal reference, not a power return.

40 V single supply. AMP_OUT rests at ~20 V DC; SPK_OUT has DC blocked by C10.
C10 positive faces amplifier; 4700u / 50V gives ~4.23 Hz into 8 ohm.
VREF = filtered divider midpoint (~20 V): input/feedback only, never speaker return.
C1 must remain fitted for a ground-referenced source; DC bypass R2 removed.
C8/C9 filter VREF: C6/C7 bypass the 40 V supply. R10 changed to preserve bias.
Startup/shutdown thump and reference settling require bench verification.

Inventory/footprint assignments: see review/inventory-2026-10-25/README.md.
Blank footprints are explicit purchase/user-assignment gaps. No PCB layout performed.

CURRENT SINK SETUP / THERMAL

RV1 wiper toward SENSE_AVG = minimum current. Start unloaded/current limited.
Set each R19/R20 to ~330 mV (1.00 A each): R21 ~540 mV (2 A total).
R25 = 470R restores adjustment with 0R33 ballasts; R21 stays 0R27.
Sense = I x (0.27 + 0.33/2): Q10-base/sense ratio ~0.52 to 1.
Range depends on Q10 temperature/beta and tolerances; max can exceed 3 A.
R27 lowers current on open wiper. Thermally couple Q10 to sink heatsink.
Reset/readjust bias after replacement and warm-up; verify each branch.
Power-device collector tabs are live: substantial heatsinking/isolation required.
Each parallel device: own SQP500 0R33/5W wirewound ballast and 10R stopper.

FIRST-PASS LIMITS / BENCH TBC

Output-stage idle supply power ~80 W: ~19-20 W per power transistor.
12 W: 13.86 Vpk / 1.73 Apk. 15 W: 15.49 Vpk / 1.94 Apk.
2 A sink gives ideal resistive Class-A current ceiling ~16 W / 8 ohm.
Upper bank peaks ~3.94 A: budget beta=35 => ~113 mA base drive.
Q8/Q9 require thermal margin: Q6/Q7 ~8 mA, ~0.16 W each at idle.
C2 100p, C5 DNP stoppers and L1: bench stability/step response TUNE.
AMP_OUT ~20 V DC; speaker output AC coupled. Test capacitor charging,
rail sag, sink transient response and hot DC SOA before PCB layout.
No simulation, bench qualification, PSU design or PCB work in this revision.

Sheet: /	
File: ClassA.kicad_sch	
Title: Mono single-ended Class-A / 2SC5198	
Size: User	Date: 2026-10-06
KiCad E.D.A. 10.0.5	**Rev: 0.2 SINGLE SUPPLY**
	Id: 1/1